

Visualizing Interchange Patterns in Massive Movement Data

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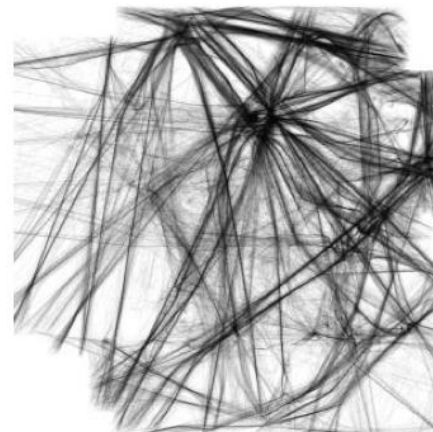
- 1. Introduction
- 2. Related Work
- 3. Our Approach: Interchange Circos Diagram
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1.1 Movement Data

- Trajectories of moving objects whose positions change over time.
- Movement data sets are huge nowadays with advances in acquisition technologies:
 - ✓ Cell phones, GPS, sensors, etc.
 - ✓ Video/image analysis
 - ✓ RFID card systems used in public transport



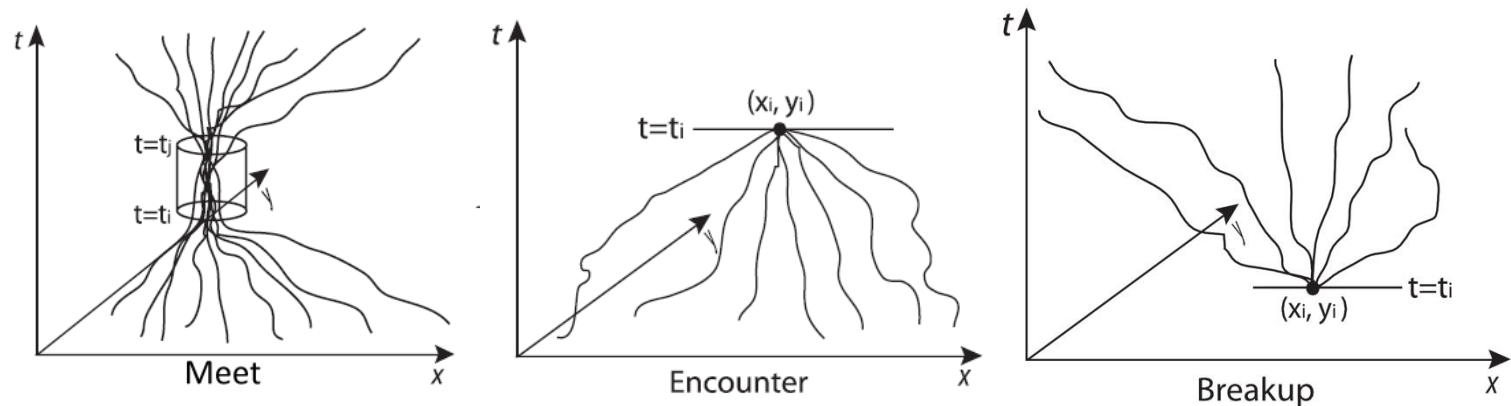
[Hurter et al. 2010]

1.1 Movement Data

- Analysis of movement data have benefited many disciplines:
 - ✓ Animal habitats study [Ganskopp et al. 2001]
 - ✓ Traffic flow reconstruction from discrete movement data [Sewall et al. 2011]
 - ✓ Function road models for traffic simulation [Wilkie et al. 2012]

1.2 Motion Patterns

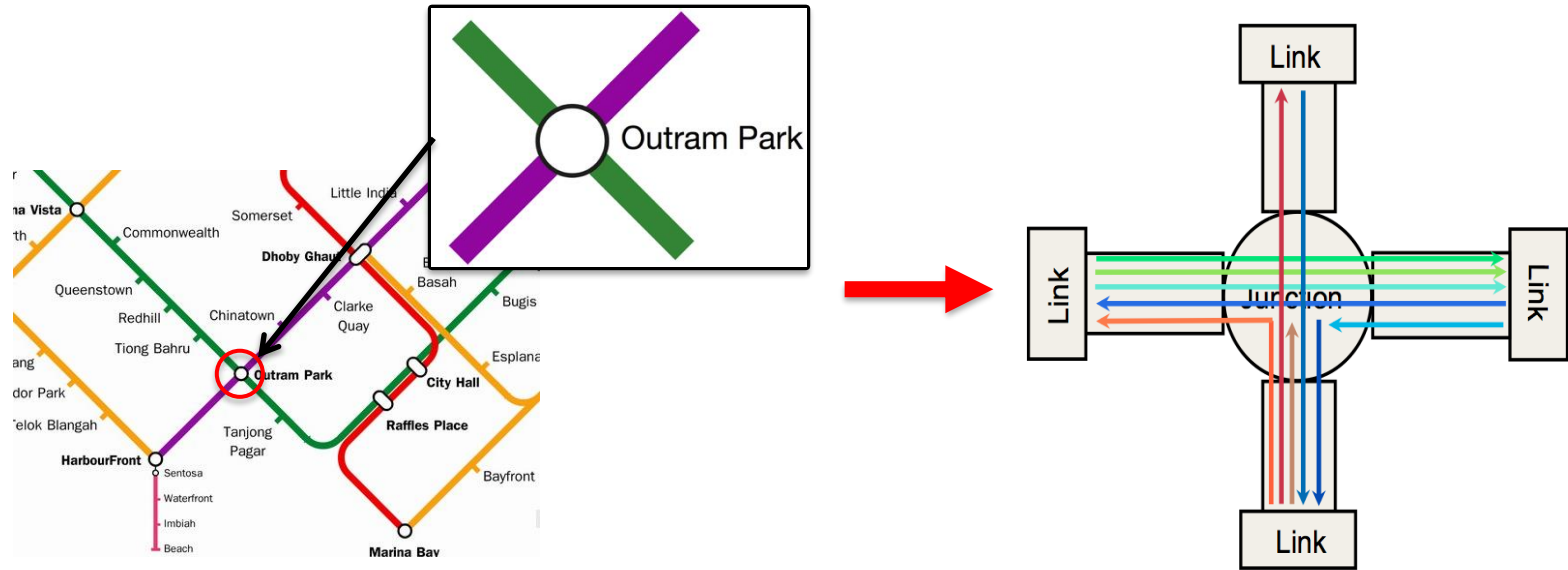
- Describe combined behaviours of a group of moving objects in space and time [Laube et al. 2005, Rinzivillo et al. 2008, Dodge et al. 2008, Liu et al. 2011].



[Dodge et al. 2008]

1.3 Interchange Pattern

- Describes how moving objects redistribute when passing through a junction node in network.



1.3 Interchange Pattern

➤ Interchange is common in our daily lives...



People interchange at crossroads

1.3 Interchange Pattern

- Interchange is common in our daily lives...



Vehicles interchange in road networks

1.4 Applications

- Visualizing interchange patterns can help understand human mobility patterns and relation with the environment.



1.4 Applications

- Visualizing interchange patterns can assist transportation planning:
 - ✓ design road intersection
 - ✓ design interior routes between station platforms



Crossroad examples: cloverleaf, traffic circles, diamond, trumpet

1.4 Applications

- Interchange patterns should be studied carefully, otherwise...



Crossroad congestion



Crowded subway station

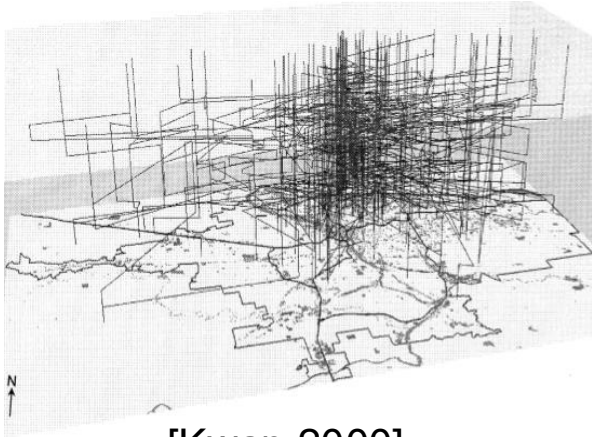
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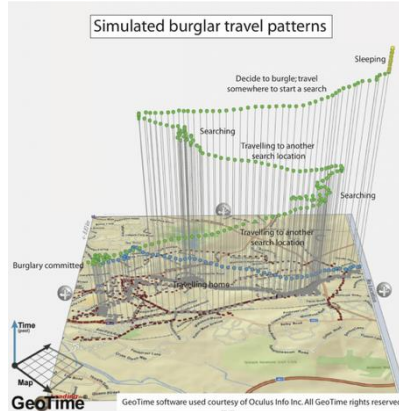
2 Movement Data Visualization

Direct
Depiction

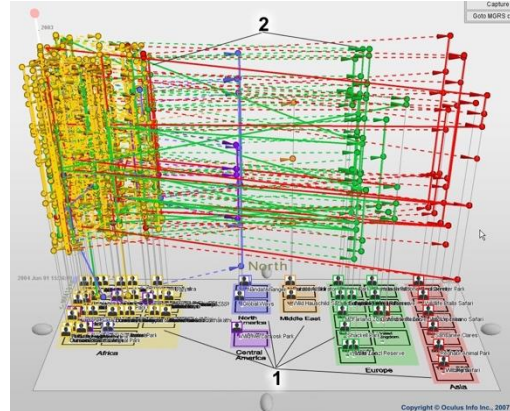
- Directly plot trajectories into 2D / 3D displays.
- Cause visual clutter and occlusion when data is large.



[Kwan 2000]



[Kapler and Wright 2004]



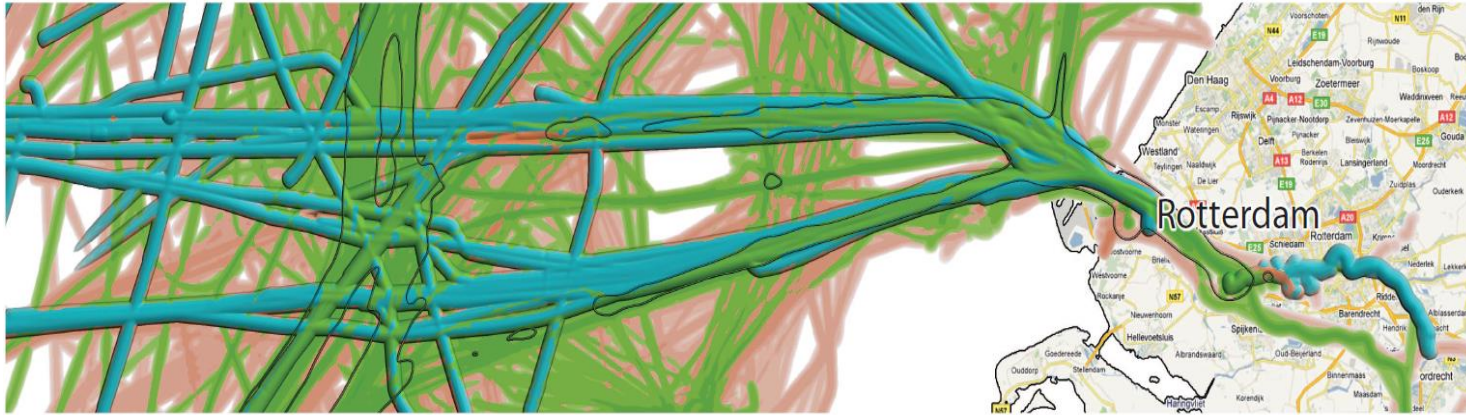
[Proulx et al. 2007]

2 Movement Data Visualization

Direct
Depiction

Density
Map

- Summarize trajectories and overview distribution.
- May not effectively show the flow direction.



[Scheepens et al. 2011]

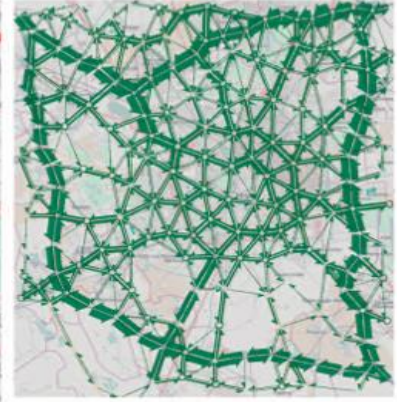
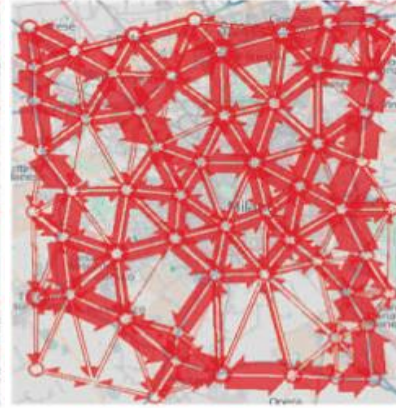
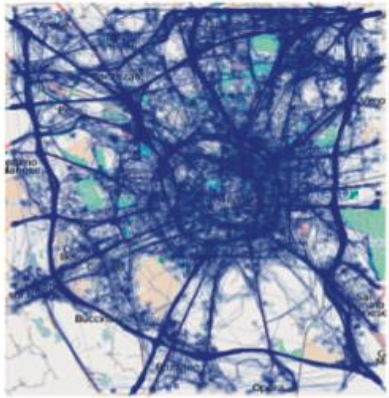
2 Movement Data Visualization

Direct
Depiction

Density
Map

Spatial
Generalization

- Partition underlying territory into appropriate areas.
- Aggregate flows between these areas and visualize it.



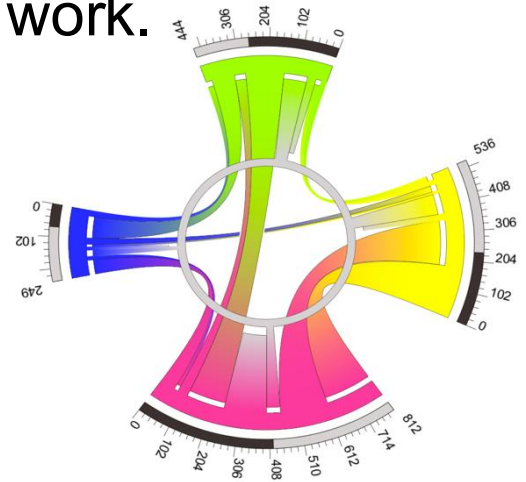
[Andrienko and Andrienko 2011]

2 Movement Data Visualization



➤ Visualize interchange patterns in movement data, which has not been studied by previous work.

- ✓ Flows volumes
- ✓ Flows directions
- ✓ Flows redistributions at junctions

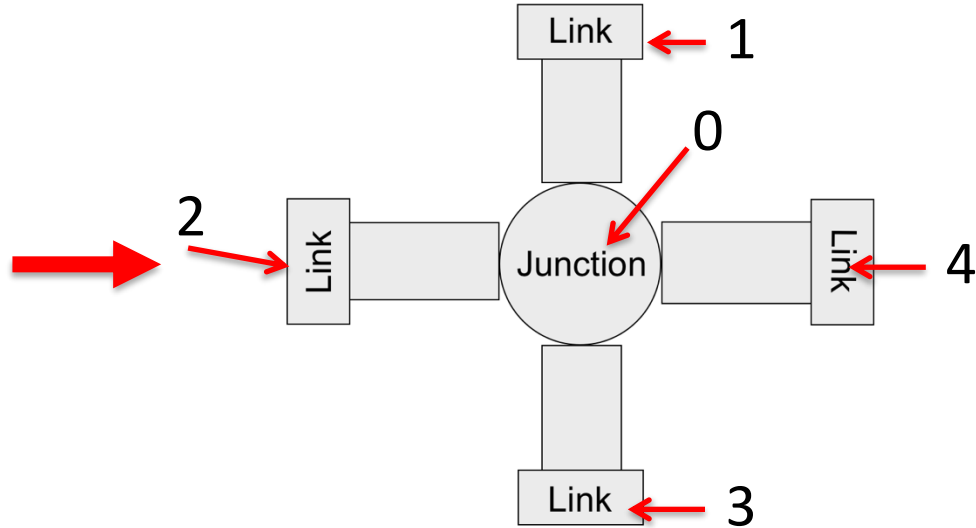
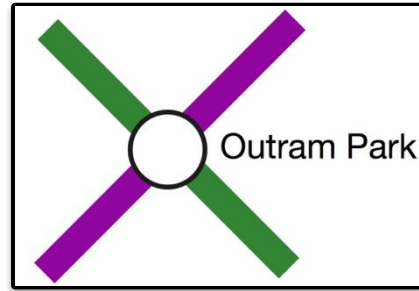


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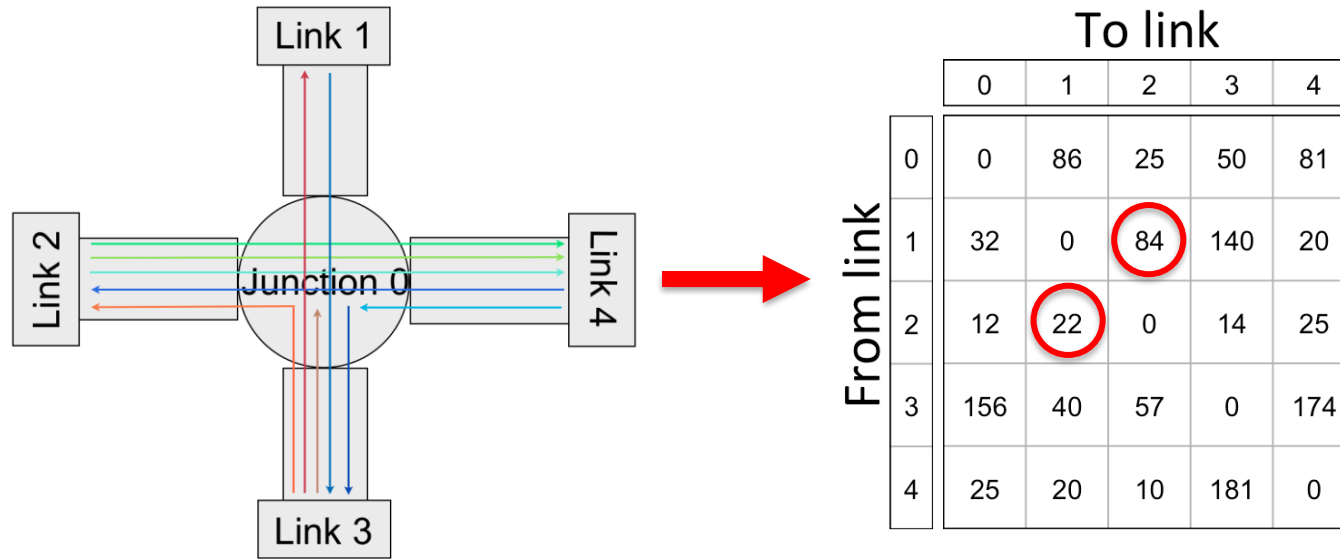
3.1 Interchange Matrix

- A network junction can be modeled as a node with n connecting links.



3.1 Interchange Matrix

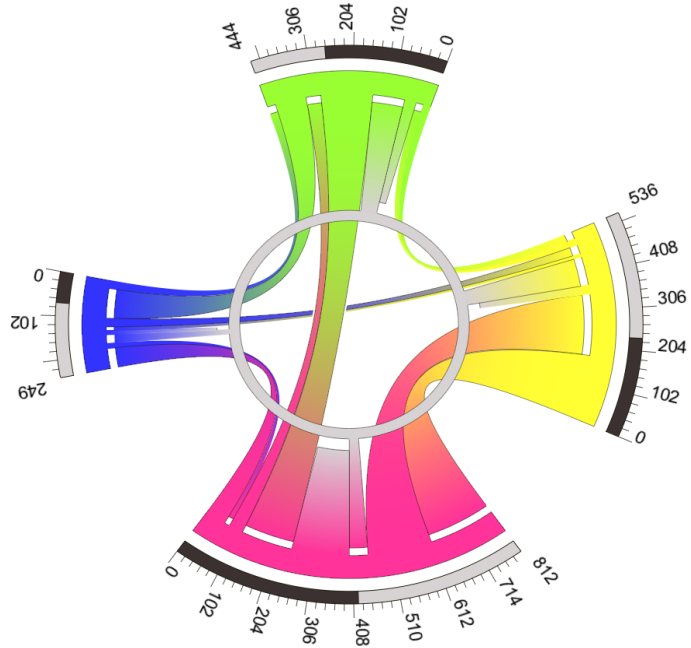
- An $(n+1)$ -by- $(n+1)$ matrix, where n is the junction's valence.



3.2 Interchange Circos Diagram

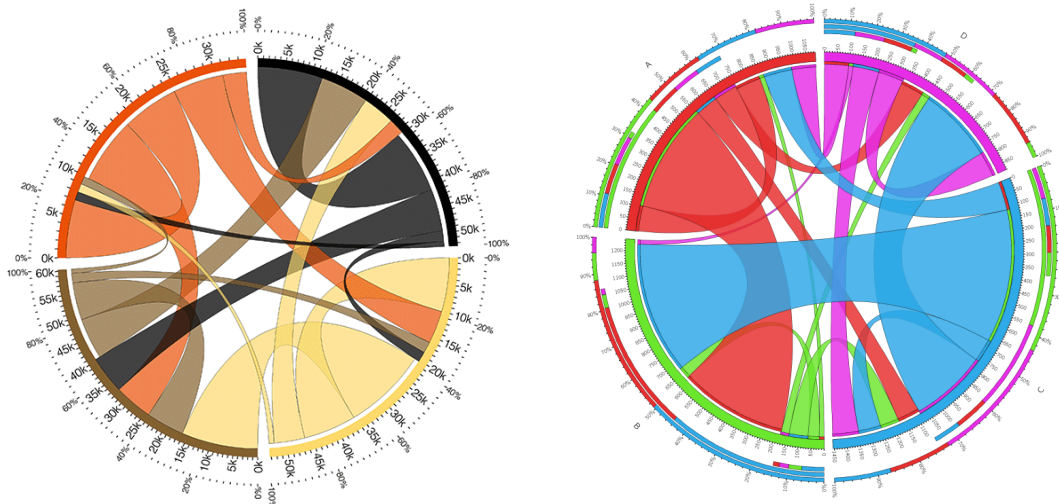
- Proposed to represent the interchange matrix.

		To link				
		0	1	2	3	4
From link	0	0	86	25	50	81
	1	32	0	84	140	20
	2	12	22	0	14	25
	3	156	40	57	0	174
	4	25	20	10	181	0



3.2 Interchange Circos Diagram

- The original circos figure examines mutual relationship, such as similarity and difference, among genomes.



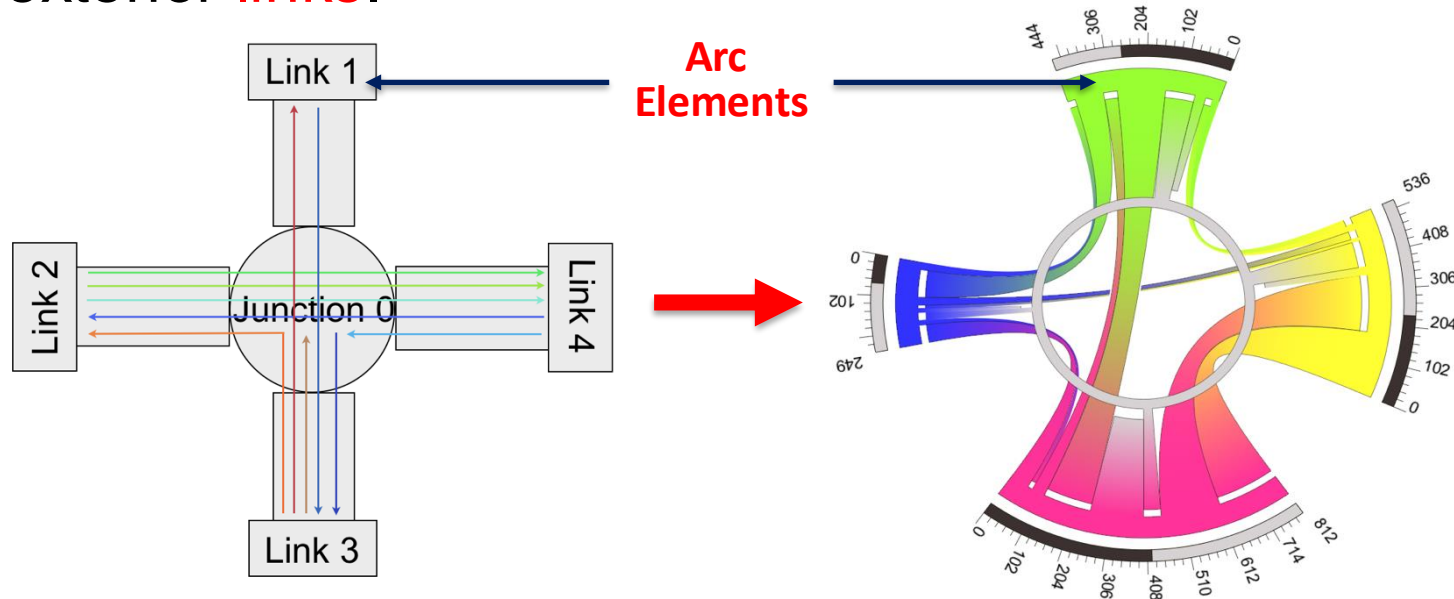
Example circos figures [Krzywinski et al. 2009]

3.3 Design Considerations

- Absolute and relative flow volumes
- Flow directions
- Ratio of total incoming and outgoing flow volumes of each link
- Flows starting / ending at the junction itself
- Correspondence to the geographical nature
- Temporal and spatial variations

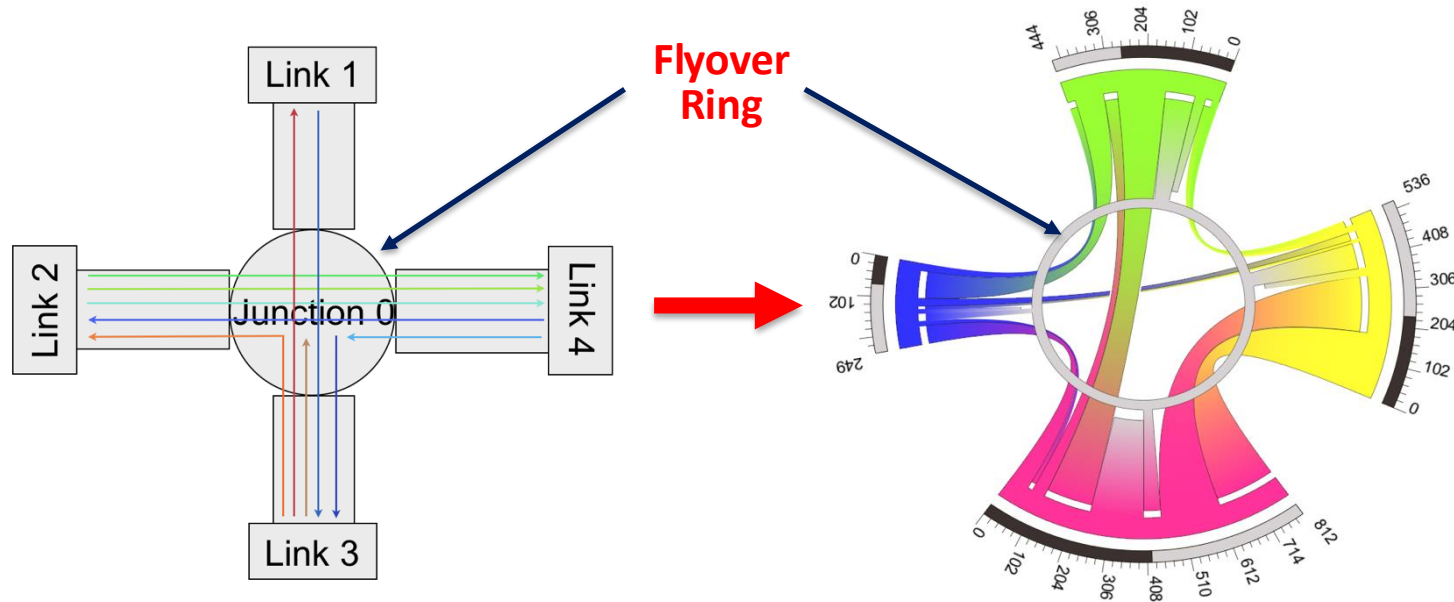
3.4 Design

- First, **arc elements** with **color labels** to represent the exterior **links**.



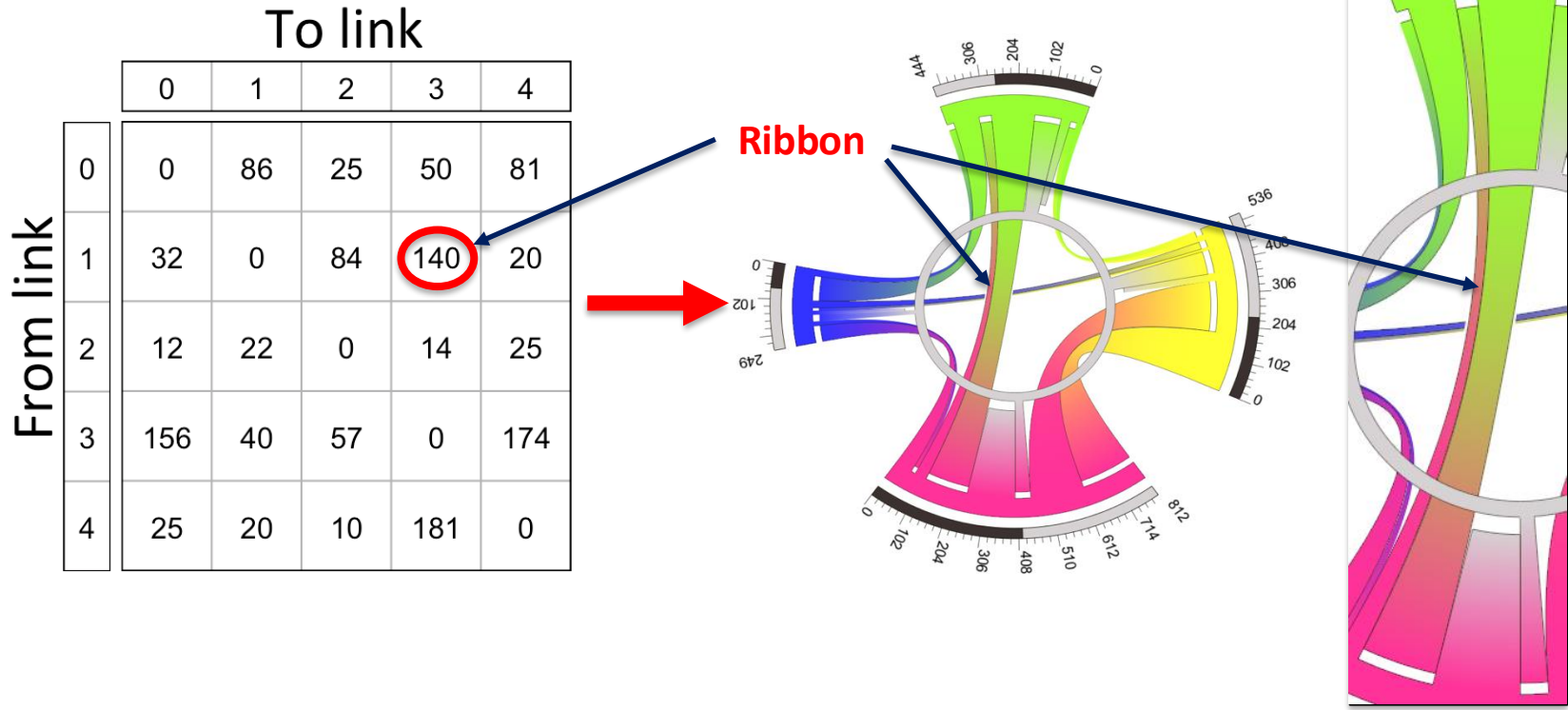
3.4 Design

- Second, a **flyover ring** to represent the **junction** itself.



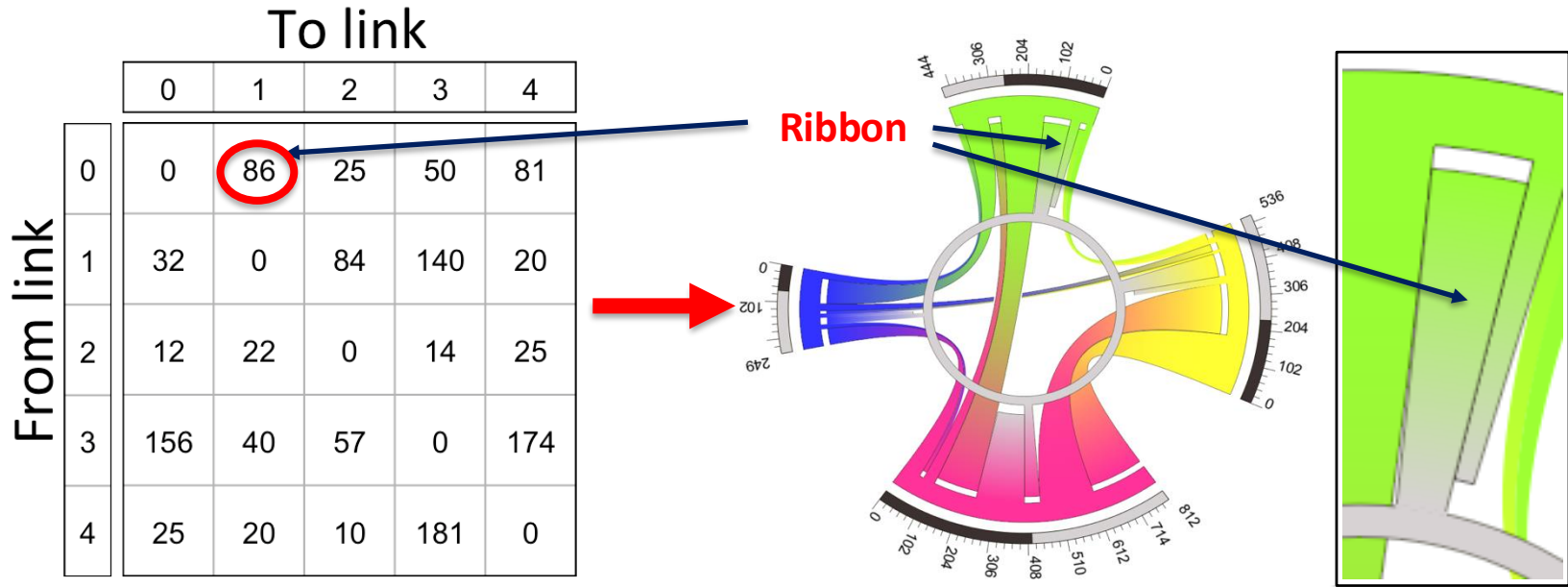
3.4 Design

- Third, connecting **ribbons** to represent **flows** in-between.



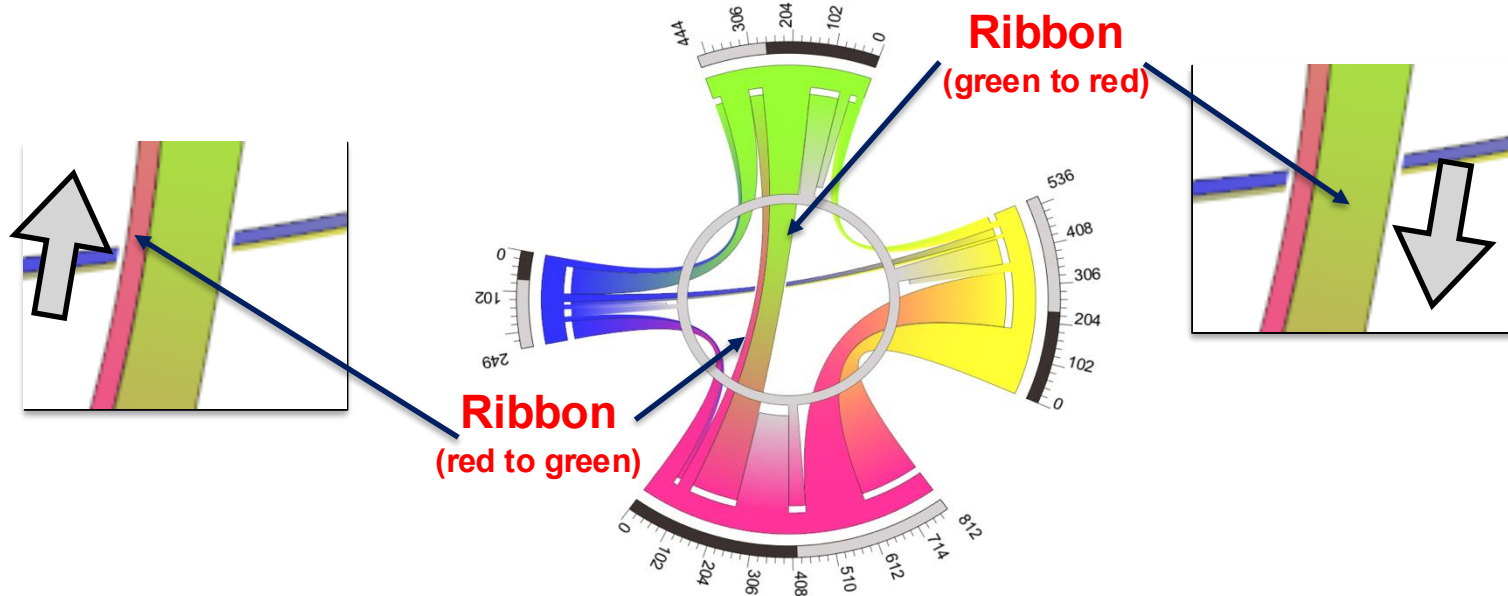
3.4 Design

- Third, connecting **ribbons** to represent **flows** in-between.



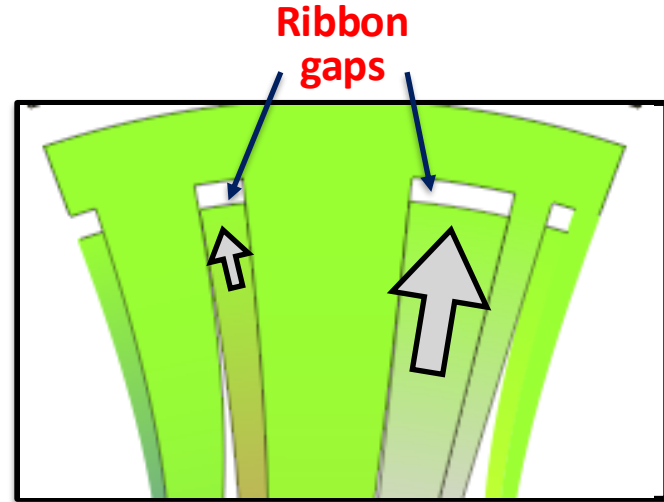
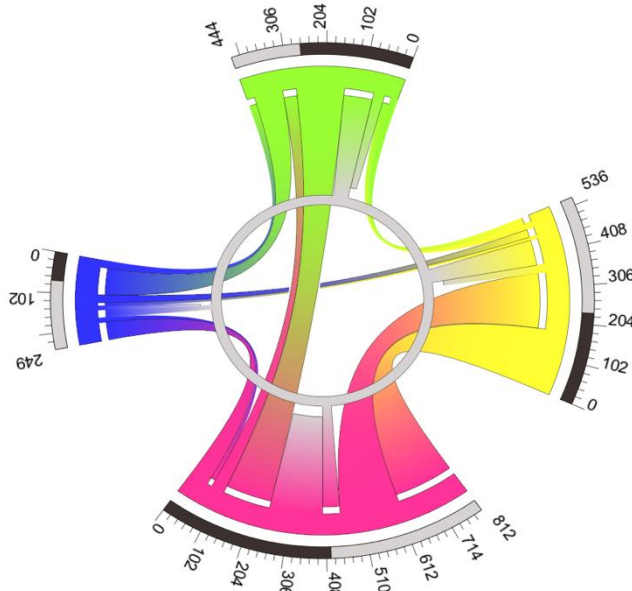
3.4 Design

- **Flow direction** is indicated by color change along each ribbon.



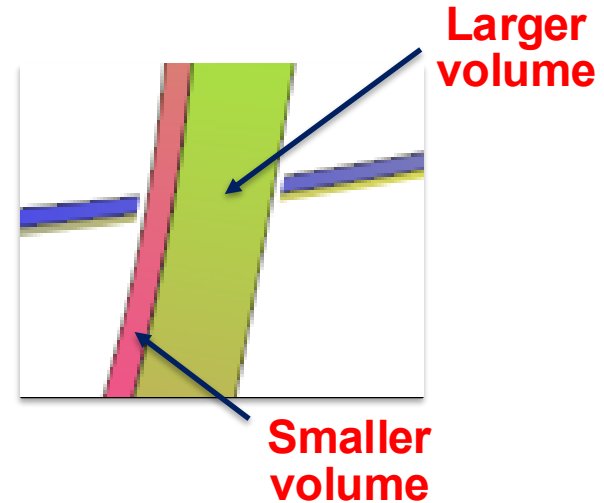
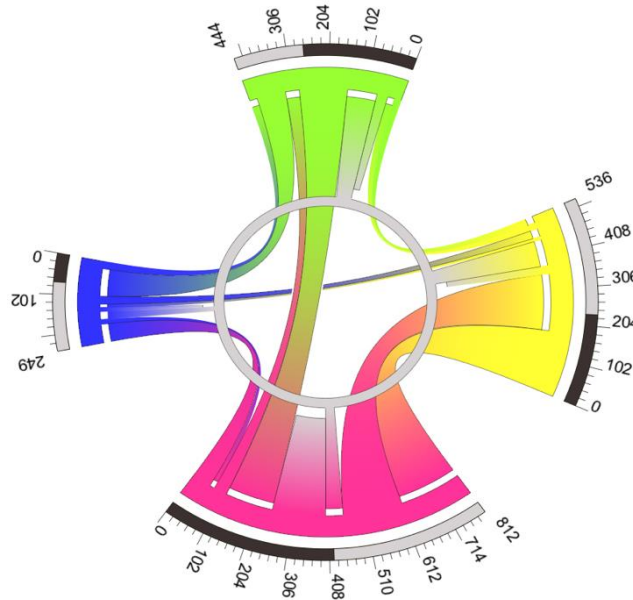
3.4 Design

- Flow direction ends with a ribbon gap.



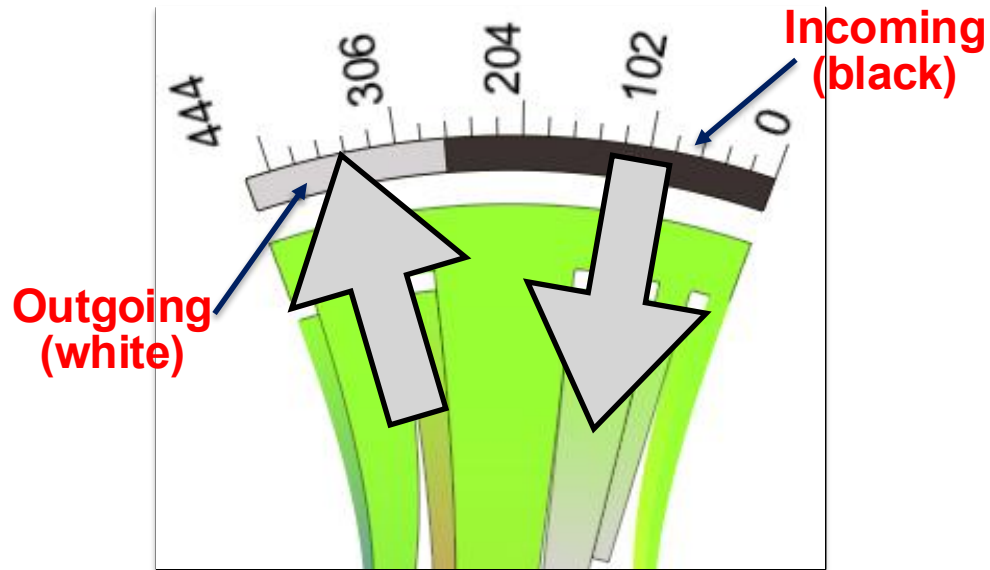
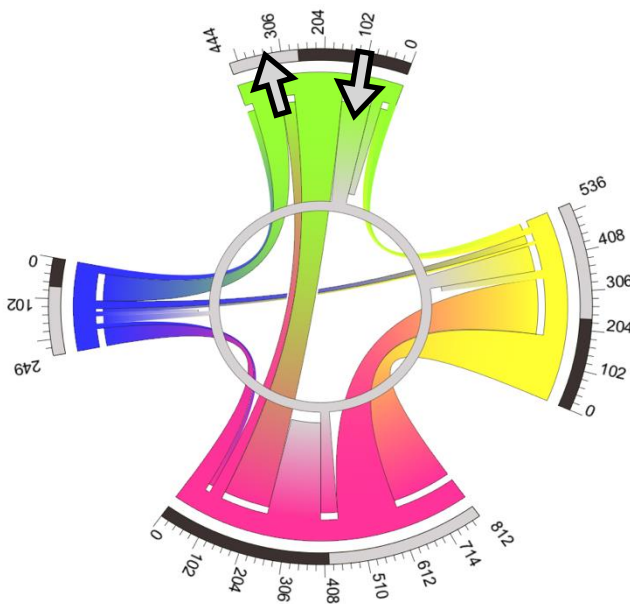
3.4 Design

- Ribbon width indicates the flow volume.



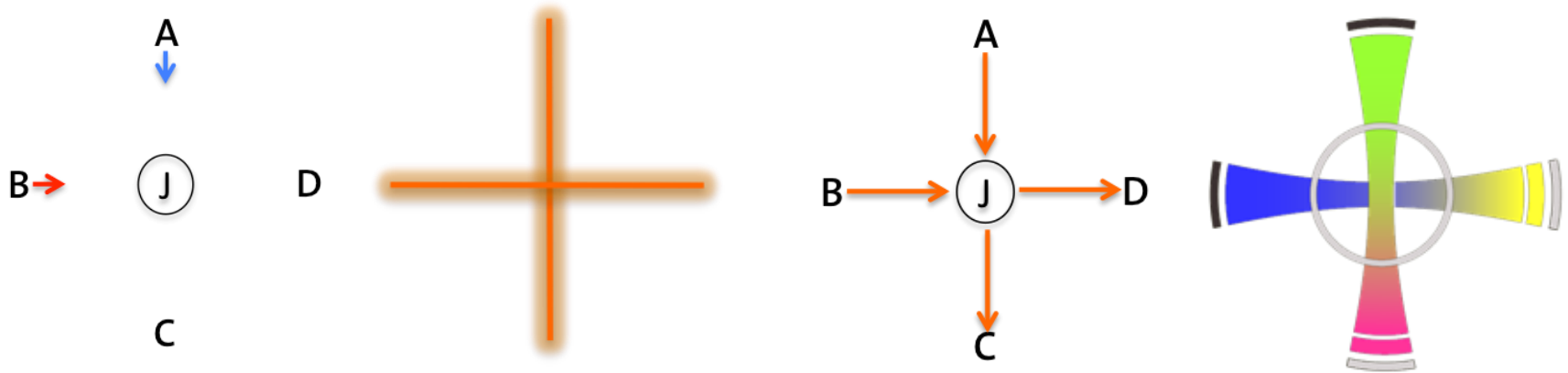
3.4 Design

- Lastly, **statistical boxes** summarizes the **flow volumes**.



3.5 Comparison with Existing Visualization

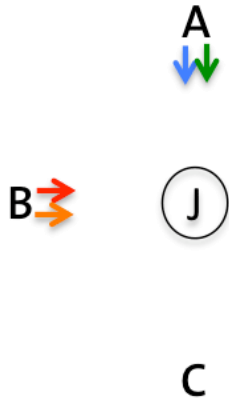
- Interchange circos diagram can present interchange patterns.



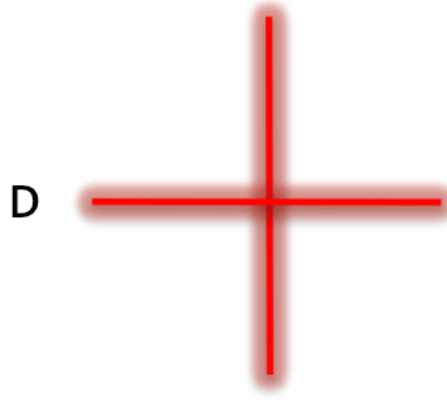
(a) raw trajectories; (b) density map; (c) spatial generalization; (d) our method

3.5 Comparison with Existing Visualization

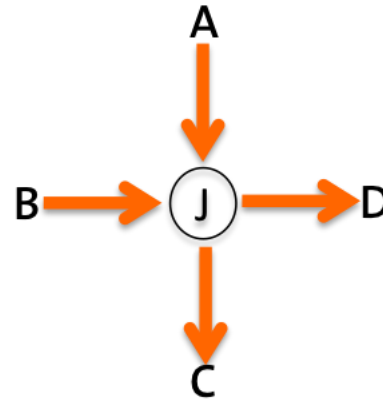
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(a) raw trajectories;



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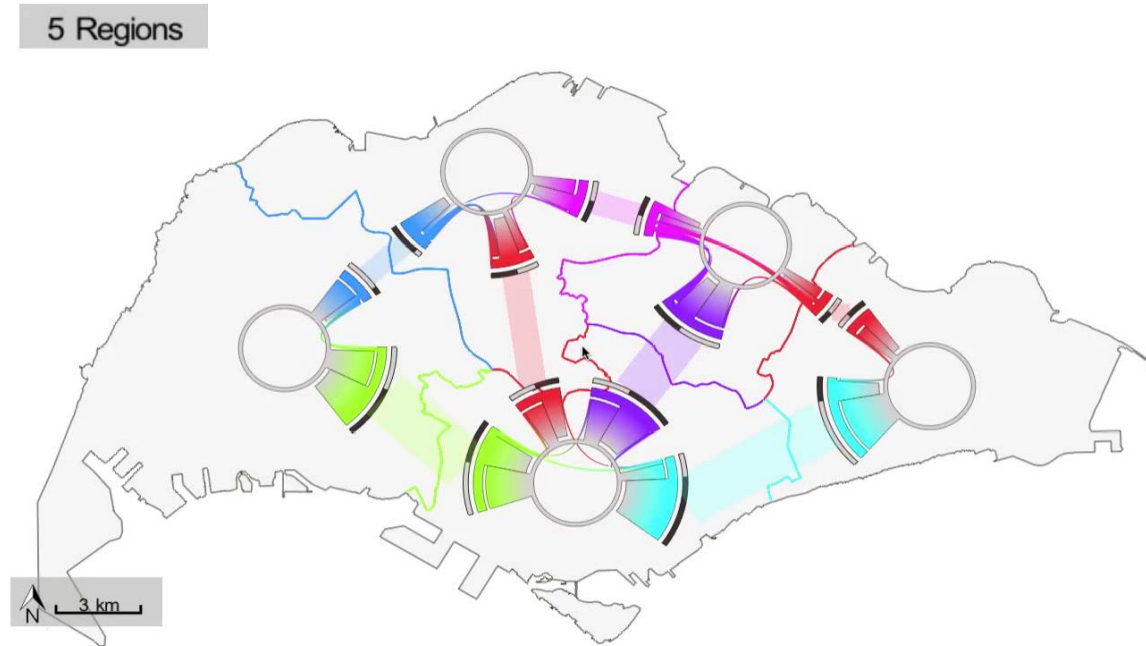
4.1 Data

- Singapore one day public transport data contains around 2 million trajectories.
- Retrieve about 1,200 crossroads from the road network.
- Load 88 subway stations from Singapore subway network.
- Generate 7 scales of zones from administration regions.

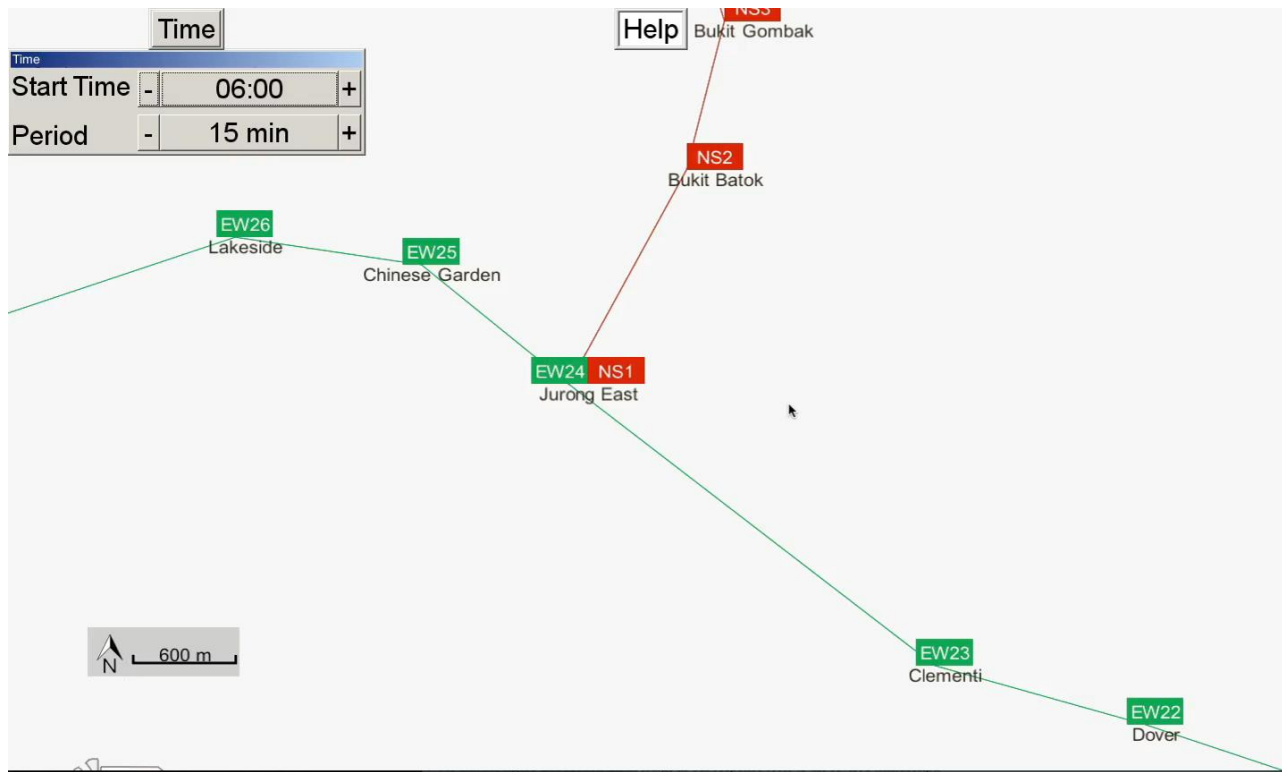
4.2 Implementation

- Intel Core i7 2 2GHz MacBook Pro with 8GB memory and an AMD Radeon HD 6490 graphics board.
- Data storage:
 - ✓ Partition time into 15 minutes, total $24 \times 4 = 96$ intervals
 - ✓ Total about 1,600 junction nodes
 - ✓ Total data size: $96 \times 1600 \times 25 \times 4 \approx 15$ MB
(average 5×5 interchange matrix)

4.3 Visualization



4.3 Visualization



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5.1 Conclusion

➤ Our contributions include:

- ✓ A formal definition of interchange patterns
- ✓ A novel visual representation called the “Interchange Circos Diagram” to present interchange patterns
- ✓ Pre-computation of interchange matrices to allow real-time interactive visual analysis of interchange patterns
- ✓ A working visual analysis system designed for studying the interchange patterns in Singapore public transport data

5.2 Future Work

- Apply interchange circos diagram to other domains, e.g., network, web and energy flow.
- Enhance our design of interchange circos diagram, particularly about presenting flow directions and volumes.
- Design interaction techniques with various types of displays, e.g., high resolution screen and mobile devices.

The End

Q & A